

ORAL PRESENTATION

Indirect Thermal-Aware Supervision for Agonist– Antagonist SMA Ankle Actuation

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Motivation and control challenge

Compact actuation is attractive, but heat recovery limits endurance

Why SMA?

Lightweight, **silent** and compact actuation for early-stage ankle mobilization.

Main limitation

Heating is fast, but **cooling is slow**; repeated activation can produce **heat accumulation**.

Control challenge

Preserve **closed-loop tracking** while reducing **unnecessary Joule heating**.

The goal is not only accurate motion, but **sustained tracking** through control decisions that reduce unnecessary heating.

Core idea

Do not only ask:

“How much should I track?”

Also ask:

When should energy enter the actuator?

Agonist-antagonist SMA

Free cooling

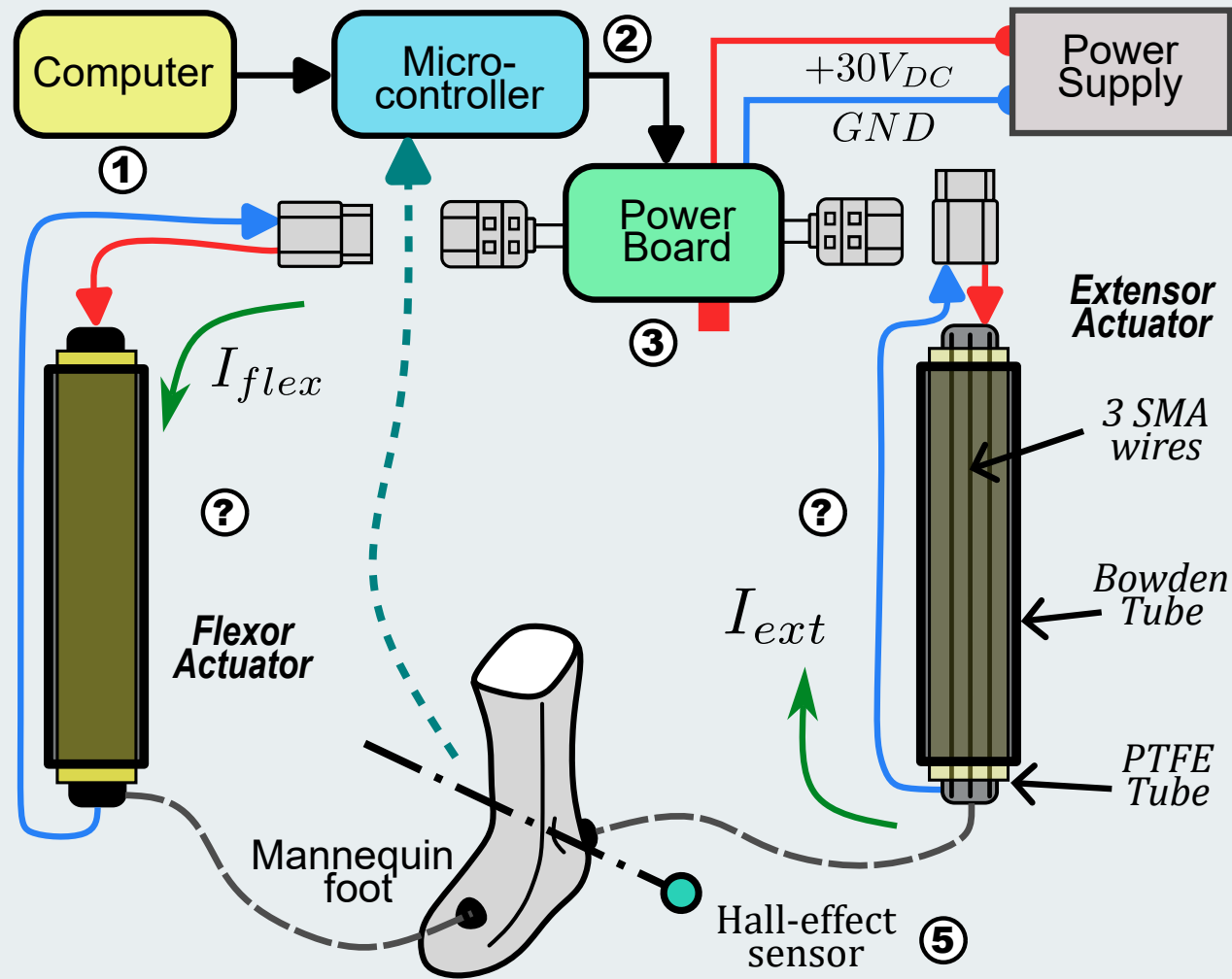
Thermal-aware supervision

BPID tracking

SYSTEM

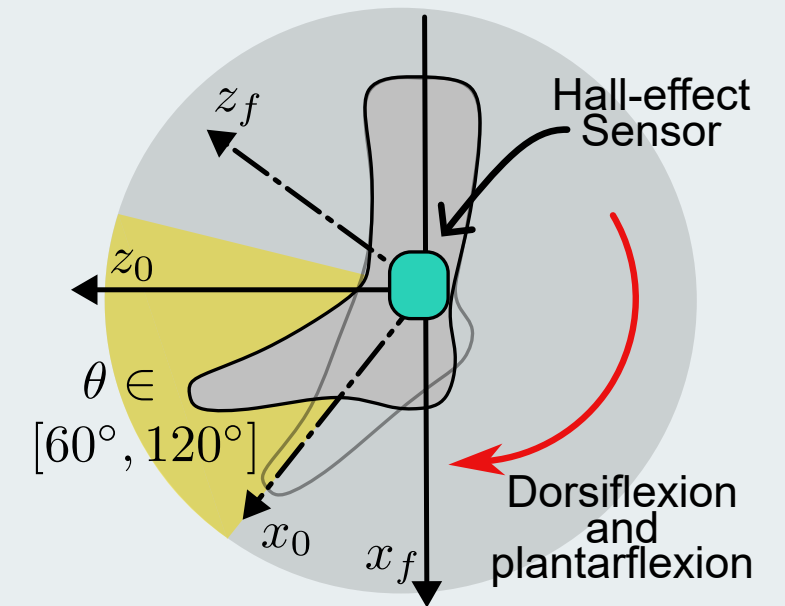
Antagonistic SMA ankle module

Actuation architecture



Hardware-in-the-loop Setup

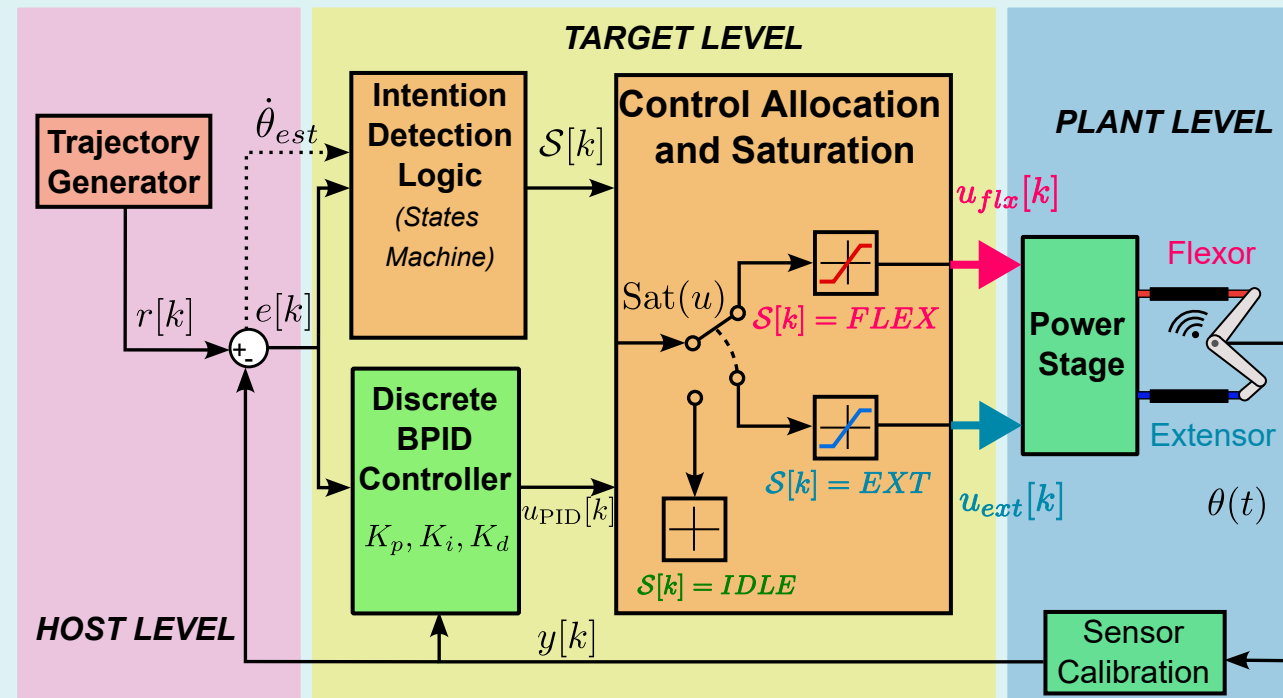
1. Host computer
2. STM32F4 target
3. Power electronics
4. Physical plant
5. Position sensor



CONTROL

Tracking + thermal-aware supervision

Closed-loop control scheme



A two-layer hierarchical logic executed across three layers.

The control scheme consists of:

Supervisory decision layer

Determines when each SMA group is allowed to receive Joule heating.

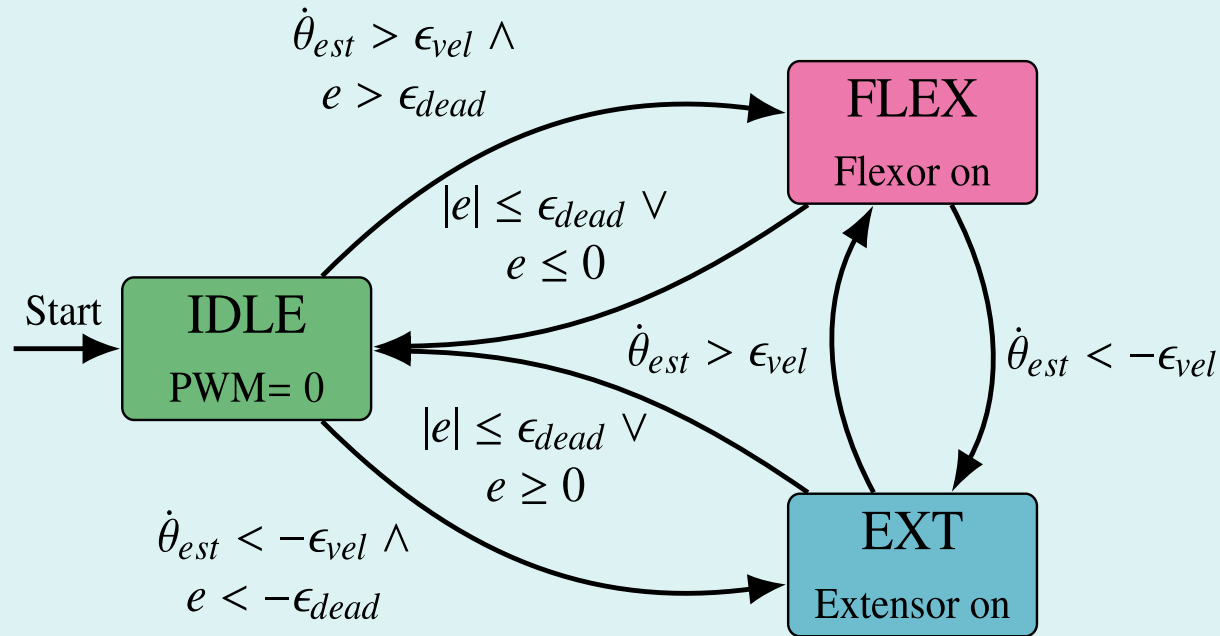
Bilinear PID layer

Computes how much control effort is applied to the active group.

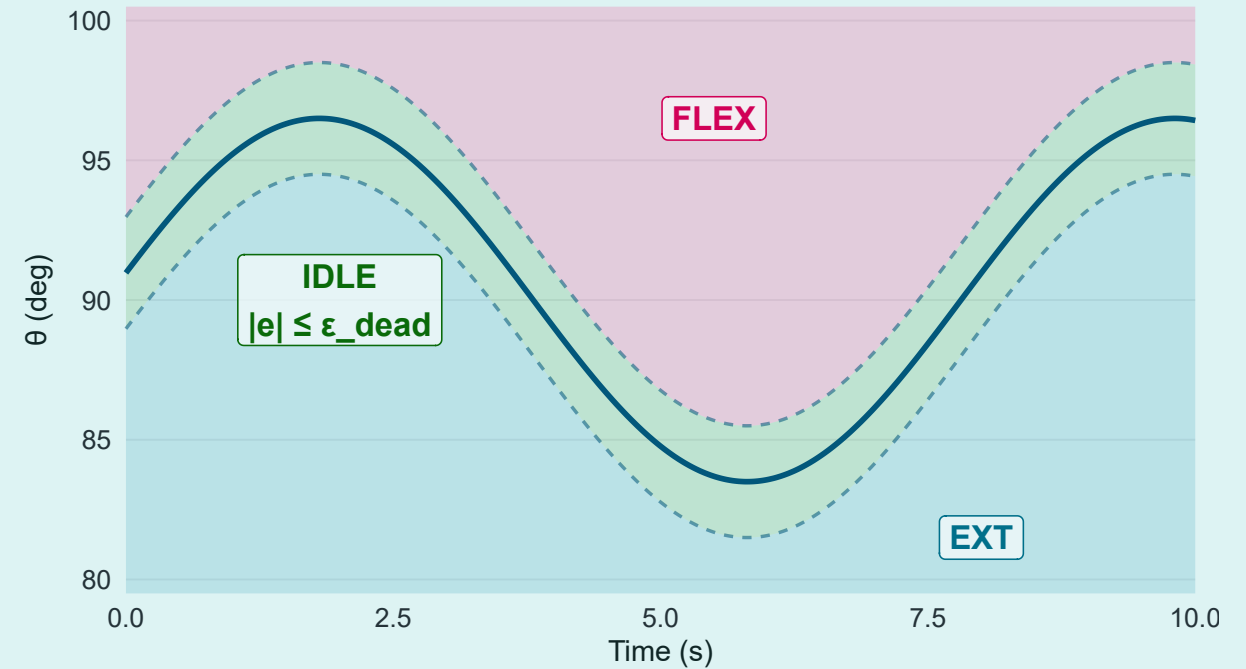
The **supervisor** decides **when to heat**; the **BPID** decides **how much**.

State logic

State machine (motion intention and tracking error)



Supervisory flowchart

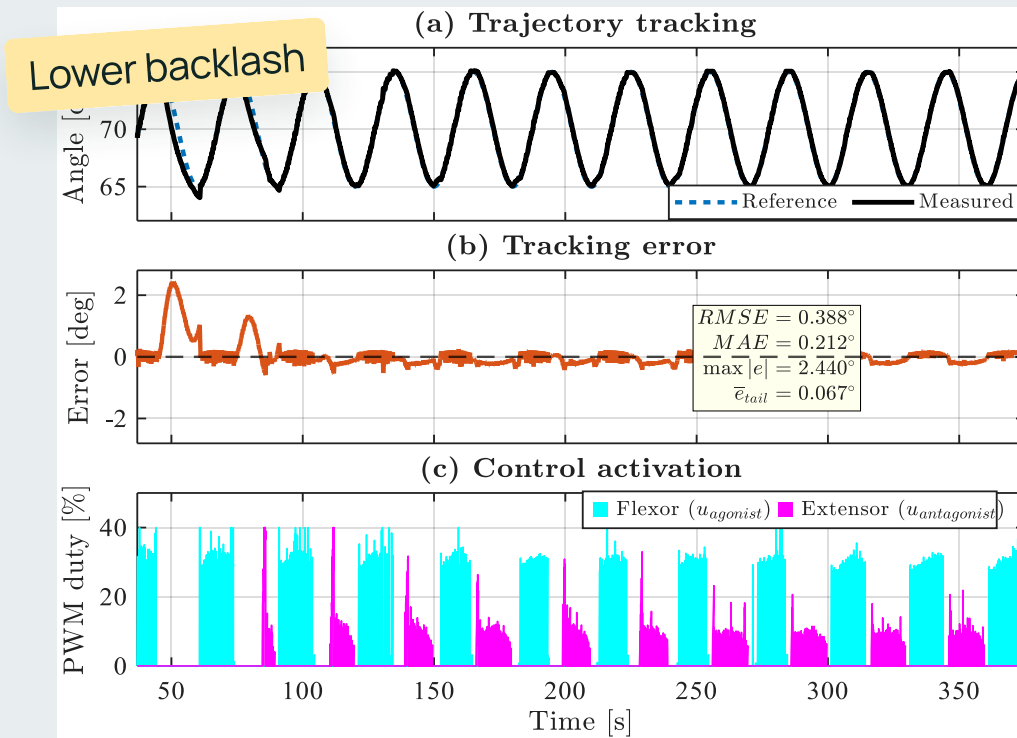


Conceptual plot of supervisory bands

Inside the band, the system stays in **IDLE** and exploits **free cooling**.

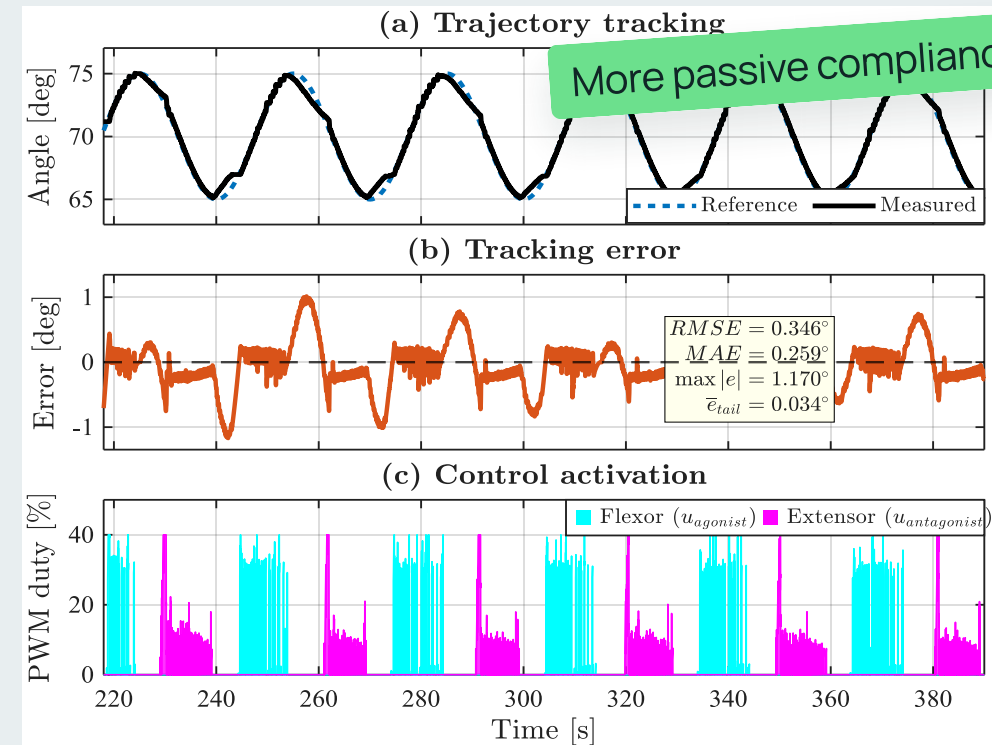
RESULTS

Results: configuration comparison



High pre-tension configuration

$$RMSE = 0.388^\circ \cdot MAE = 0.212^\circ \cdot \max |e| = 2.440^\circ$$



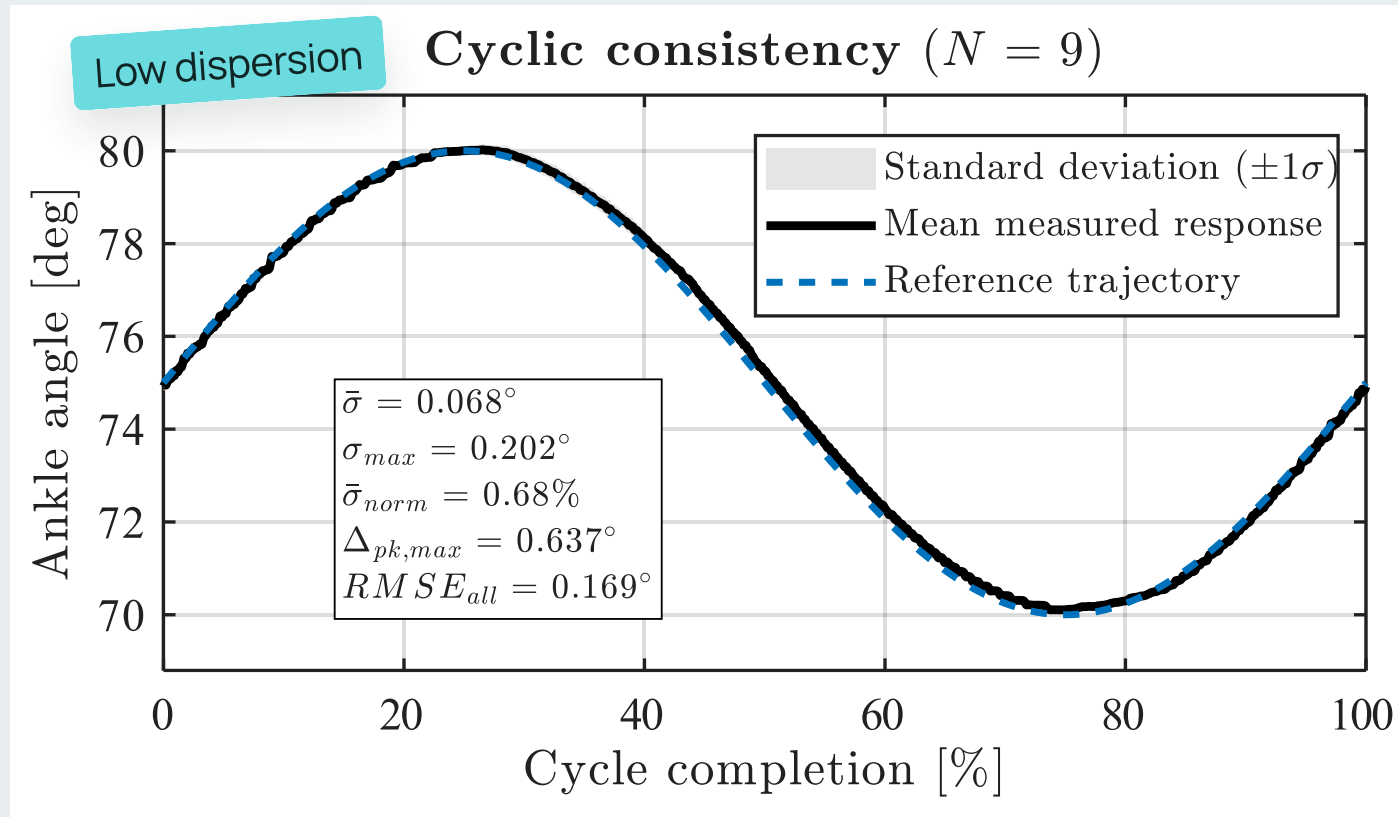
Relaxed initial state configuration

$$RMSE = 0.346^\circ \cdot MAE = 0.259^\circ \cdot \max |e| = 1.170^\circ$$

High pre-tension improves early mechanical response, but the **relaxed state** reduces peak error and promotes recurrent **free-cooling intervals**.

Results: cyclic repeatability

The response remains consistent from cycle to cycle



Mean dispersion: $\bar{\sigma} = 0.068^\circ$

Maximum dispersion: $\sigma_{max} = 0.202^\circ$

Peak mismatch: $\Delta_{pk,max} = 0.637^\circ$

Overall cyclic RMSE: $RMSE_{all} = 0.169^\circ$

This indicates:

High repeatability

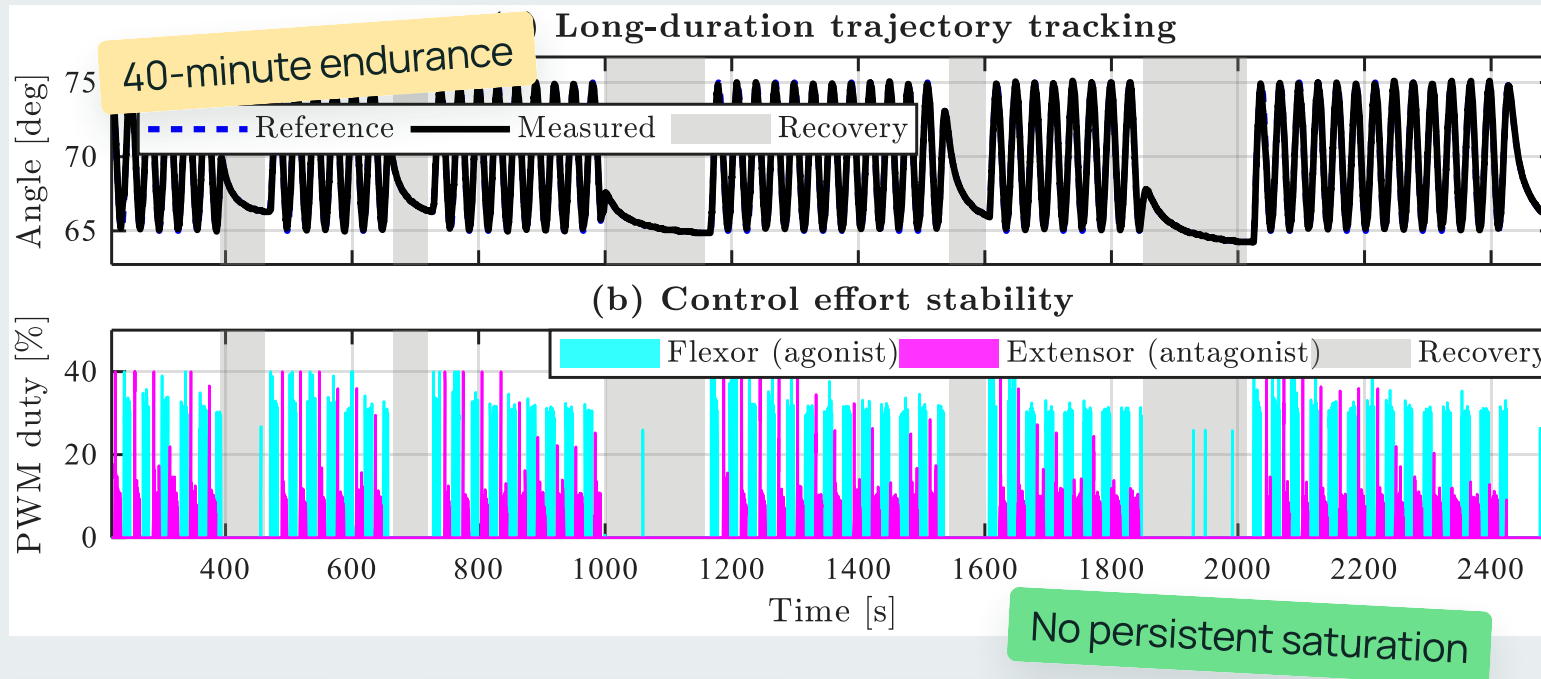
Low inter-cycle drift

Stable closed-loop behavior

The **small dispersion band** shows that tracking is not only accurate, but **repeatable cycle after cycle**.

Results: long-duration endurance

Stable operation over 40 minutes without thermal saturation



Duration

40 min closed-loop operation

Tracking accuracy

$RMSE_{active} = 0.218^\circ$

Worst-case error

$\max |e| = 1.384^\circ$

Thermal behavior

Recovery intervals prevent persistent heat accumulation

Recovery intervals

No tracking collapse

Closed-loop recovery

The key result is **sustained tracking**: recovery windows are embedded in the control strategy, preventing persistent heat accumulation while keeping the loop active.

Discussion and take-home message

What do these results mean?

Tracking accuracy alone is not enough: in SMA systems, **thermal recovery** must be part of the control architecture.

The supervisor does not need direct temperature sensing. It uses **motion intention** and **signed tracking error**.

Inside the band, the controller stays in **IDLE** and exploits **free cooling**.

Take-home message

Indirect thermal-aware supervision enables SMA ankle actuation that is **and thermally robust**.

Thank you

DISCUSSION

Questions?

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